E2E report — lab from-start run (2026-10-25)

When: 2026-10-25 04:51–04:56 UTC (10:21–10:26 IST) **Where:** this Mac (compile) + dh2 `172.22.1.2` (run) **Repo:** `opi-nvidia-dpf-adapter` **Image on lab:** `controller:lcm-20261024b` **Job dir:** `/tmp/lab-job-lcm-from-start/` (deleted after)

Verdict: LCM e2e on dh2 is **green**. User-flow **stalls at DPF provision** (DPU Rebooting). Dataplane is **not** ready. That is DPF/hostagent, not an LCM install fail.

Not run: `mlxfwreset`, host reboot, helm-uninstall, AMD stubs, lab `/etc` changes.

1. Scoreboard

Surface	Result
VPN F5 Access	Connected
ping 172.22.1.2	0% loss, ~323 ms
SSH dh2	hostname `dh2`, k3s `v1.36.4+k3s1`
Mac `go test ./pkg/lifecycle/`	**ok 1.291s**
Mac `go test -race ./pkg/lifecycle/`	**ok 3.847s**
`make govulncheck`	**exit 0** — No vulnerabilities found
linux/amd64 `lifecycle.test`	ELF 64-bit, 40 808 912 bytes
dh2 compiled suite	**188 PASS / 0 FAIL / 0 SKIP**, 253 RUN, **EXIT 0**
Live LCM conditions	Supported=True, VendorOperatorReady=True
Helm pin claim	`dpf-operator` v1 deployed, `mode=adopted`
ClusterRole wildcards	**NO_WILDCARDS**
Secret watch errors	**0**
AMD `dscdevices`	**NotFound** (expected on BF2)
DPU `dh2-bluefield-2`	**Rebooting** / SystemLevelReset
Dataplane DPUServices	4 Success / 6 Pending
Job dir after	deleted

2. Commands that were run

2.1 Hop

```
scutil --nc start "F5 Access"
ping -c 2 172.22.1.2
ssh -o IdentityAgent=none -o IdentitiesOnly=yes \
  -o IdentityFile=$HOME/.ssh/opi_lab_key -o UseKeychain=yes \
  -o KexAlgorithms=curve25519-sha256 \
  root@172.22.1.2 'hostname && nproc && free -h && k3s --version'
```

dh2: 64 CPUs, 247 Gi RAM (7.2 Gi used), Ubuntu 26.04.1, kernel 7.0.0-38-generic, containerd 2.3.4-k3s1. Node `dh2` Ready, INTERNAL-IP `172.22.1.2`.

2.2 Compile on this Mac (dh2 has no usable `go test` path)

```
cd /Users/kartikey0104/Desktop/opi-dpf-adapter/opi-nvidia-dpf-adapter
export GOTOOLCHAIN=go1.26.10
go test ./pkg/lifecycle/ -count=1 -timeout 4m          # ok 1.291s
go test -race ./pkg/lifecycle/ -count=1 -timeout 4m    # ok 3.847s
```

```
make govulncheck                                    # No vulnerabilities found
CGO_ENABLED=0 GOOS=linux GOARCH=amd64 GOTOOLCHAIN=go1.26.10 \
  go test -c -o /tmp/lifecycle.test ./pkg/lifecycle/
```

Copied into `/tmp/lab-job-lcm-from-start/` on dh2: binary, `go.mod`/`go.sum`, `config/{rbac,versions,apf,webhook}`, `cmd/main.go`, `pkg/lifecycle/*.go`, probe chart testdata.

2.3 Compiled suite on dh2

```
cd /tmp/lab-job-lcm-from-start/pkg/lifecycle
/tmp/lab-job-lcm-from-start/lifecycle.test \
  -test.count=1 -test.timeout=8m -test.v
```

EXIT 0. 188 `--- PASS`, 0 `--- FAIL`. Includes Helm claim, ClusterHelm lock/retry, OLM observe, NVIDIA pin, webhook admission, DPU observe, RBAC no-wildcard, govuln pins.

Then `rm -rf /tmp/lab-job-lcm-from-start`.

3. Live user-flow e2e (the real BlueField path)

Step	What we checked	Result
1 Enable NVIDIA	`spec.vendor=nvidia`, annotations `vendor=nvidia`, `vendor-operator-version=v26.4.0`, `adopt-existing=true`	**PASS**
2 Preflight	k3s 1.36.4; cert-manager 1/1; Argo server 1/1; NFD; maintenance-operator; DPF controllers 1/1	**PASS**
3 Helm claim	Secret `sh.helm.release.v1.dpf-operator.v1` `owner=helm` `mode=adopted` `lifecycle.opi.nvidia.com/owner=opi-nvidia-dpf-adapter` `status=deployed` `version=1`	**PASS**
4 LCM Ready	last log: `action=noop` `installed=v26.4.0` `skew=Supported` `ready=true` `operator=Ready` `kubernetes=v1.36.4+k3s1`	**PASS**
5 DPF provision	DPU phase **Rebooting**; annotation `host-power-cycle-required=true`; `rebootMethod=SystemLevelReset`; pending NVCONFIG `PF_SF_BAR_SIZE`, `PF_TOTAL_SF`; BFB `bf-bundle-340` **Ready**	**STALL**
6 Dataplane	see table below	**NOT READY**

`VendorDPUReady` is **absent** on the live object. The observer landed after image `lcm-20261024b`. `VendorOperatorReady` stays True, which is the intended split: operator healthy, DPU still rebooting.

4. Live objects

DpuOperatorConfig

- name `dpu-operator-config`, generation 2
- finalizer `lifecycle.opi.nvidia.com/cleanup`
- `VendorOperatorSupported=True/Supported` — nvidia operator v26.4.0 in `[v26.4.0, v27.0.0)`
- `VendorOperatorReady=True/Ready` — DPF CRDs, controllers, and prerequisites are healthy

Adapter

- ns `opi-nvidia-dpf-adapter-system`
- deploy image `controller:lcm-20261024b`
- pod `…-65b7466b6d-x5qpr` 1/1 Running, 0 restarts, age 15h
- leader elected; LCM workers=1
- `secrets is forbidden` count in last 200 log lines: **0**
- VSP pod `opi-nvidia-dpf-adapter-nvidia-vsp-bfhws` 1/1

DPF controllers (healthy)

`dpf-operator-controller-manager` 1/1 (image `nvcr.io/nvidia/doca/dpf-system:v26.4.0`), provisioning 2/2, dpuservice 1/1, static-cm 1/1, cert-manager 3/3, Argo server/repo/redis/application-controller 1/1.

DPU / BFB / OPI DPU

- `provisioning.dpu.nvidia.com/DPU` `dh2-bluefield-2` phase **Rebooting**
- device `mt2321xz0kvx`, BFB `bf-bundle-340` phase **Ready** (38h)
- OPI `DataProcessingUnit` `dh2-bluefield-2` product BlueField-2, node dh2 (STATUS empty — DPFReady still false)

DPUServices (~39h)

Name	Ready	Phase
multus	True	Success
nvidia-k8s-ipam-node	True	Success
ovs-cni	True	Success
servicechainset-rbac-and-crds	True	Success
cni-installer	False	Pending
flannel	False	Pending
nvidia-k8s-ipam-52111ffde9	False	Pending
servicechainset-controller-52111ffde9	False	Pending
sfc-controller	False	Pending
sriov-device-plugin	False	Pending

CrashLoop / Error (host-side, waiting on DPU):

- `dpu-cluster-1-cni-installer-js6kq` CrashLoopBackOff, exit 255, ~454 restarts
- `dpu-cluster-1-sfc-controller-node-ds-tm6mh` CrashLoopBackOff, exit 1, ~443 restarts
- `kube-flannel-ds-cw7ww` Error, ~453 restarts
- `kube-sriov-device-plugin-zc522` Pending, no node

These do **not** flip LCM to Degraded. That is correct: LCM owns operator install/observe, not DPF reboot completion.

5. What this e2e proves and does not prove

Proved on lab this run

- Hop + compiled-from-start LCM unit/integration on dh2 (188 PASS)
- NVIDIA pin is claimed and idempotent (`noop`)
- RBAC has no `*`; Secret informer disable holds (0 watch errors)
- AMD CRDs stay absent on BlueField
- Govuln pins still clean on this Mac

Not proved / still open

- Live `VendorDPUReady` on the CR (needs a new image roll)
- DPU leaving Rebooting (needs host power cycle or DPF-approved reset — **not** run)
- Kind e2e 6/6 was **not** re-run this pass (last recorded: 6/6 after docker-preflight)
- Production NGC install/upgrade of release name `dpf-operator` is still refused by `ClusterHelmClient`

6. Pass / fail you should treat as the e2e gate

Gate	This run
`lifecycle.test` on dh2 exit 0	**PASS**
One Helm Secret, claimed, no second `dpf-operator` release	**PASS**
ClusterRole no `*`	**PASS**
Supported True + VendorOperatorReady True	**PASS**
DPU Rebooting reported; no reset run	**PASS (observe)**
`dscdevices` NotFound	**PASS**
Job dir deleted	**PASS**
Dataplane Ready	**FAIL (known stall)**